Preliminaries

Notation $P_d[x]$ stands for the whole real coefficient polynomials with order d and n variables (i.e. $x \in \mathbb{R}^n$), and notation Γ_d stands for all n^{th} dimensional non-negative vectors whose summation are equivalent or smaller than d as follows:

$$\Gamma_d = \{\alpha \in \mathbb{Z}_+^n |\ |\alpha| = \alpha_1 + \alpha_2 + .. + \alpha_n \leq d\} \tag{1}$$

For $x \in \mathbb{R}^n$, notation $x^{\{d\}}$ represents the vector of all monomials of vector x whose degree is equivalent or smaller than d. Using this notation, any arbitrary polynomial of $P_d[x]$ such as $\boldsymbol{p}(x) = \sum_{\alpha \in \Gamma_d} p_\alpha x^\alpha$ can be represented by $\boldsymbol{p}(x) = x^{\{d\}^T} p$ where $p \in \mathbb{R}^{|\Gamma_d|}$ contains coefficients of $\boldsymbol{p}(x)$ such that p_α is the coefficient of monomial x^α for all $\alpha \in \Gamma_d$.

The moment matrix denoted by $M_d(y) \in \mathbb{R}^{|\Gamma_d| \times |\Gamma_d|}$ for $y \in \mathbb{R}^{|\Gamma_{2d}|}$ is a symmetric matrix with the following definition:

$$M_d(y) = \sum_{\alpha \in \Gamma_d} \sum_{\beta \in \Gamma_d} y_{\alpha+\beta} E_{d,\alpha,\beta} \tag{2}$$

where $E_{\alpha,\beta} \in \mathbb{R}^{|\Gamma_d| \times |\Gamma_d|}$ is a symmetric matrix with the following definitions:

$$\forall \alpha, \beta \in \Gamma_d:\ \ E_{d,\alpha,\beta} = \begin{cases} e_{d,\alpha} e_{d,\beta}^T, & \alpha = \beta \\ e_{d,\alpha} e_{d,\beta}^T + e_{d,\beta} e_{d,\alpha}^T, & \alpha \neq \beta \end{cases} \tag{3}$$

In (3), notation $e_{d,\gamma} \in \mathbb{R}^{|\Gamma_d|}$ is a real vector whose entries are zero except the one locating on the index of γ in set Γ_d. According to (2-3), we have:

$$M_d(y) = \sum_{\gamma \in \Gamma_{2d}} \sum_{\substack{\alpha \in \Gamma_d \\ \gamma - \alpha \in \Gamma_d}} y_\gamma E_{d,\alpha,\gamma-\alpha} = \sum_{\gamma \in \Gamma_{2d}} y_\gamma \bar{E}_{d,\gamma} \tag{4}$$

$$\forall \gamma \in \Gamma_{2d}:\ \ \bar{E}_{d,\gamma} = \sum_{\substack{\alpha \in \Gamma_d \\ \gamma - \alpha \in \Gamma_d}} E_{d,\alpha,\gamma-\alpha} \tag{5}$$

Notation $[y]_d$ stands for given vector $y \in \mathbb{R}^{|\Gamma_{d'}|}$ (let $d' \geq d$) is the truncated (projected) part of this vector such that $\bar{y} = [y]_d$ implies $\bar{y}_\alpha = y_\alpha$ for all $\alpha \in \Gamma_d$.

Finally, suppose $\boldsymbol{s}(x) \in P_{d_s}[x]$ considering $d_s < d$ that enables us to define $M_d(s, y) \in \mathbb{R}^{|\Gamma_d| \times |\Gamma_d|}$ as follows for $y \in \mathbb{R}^{|\Gamma_{2d+d_s}|}$:

$$M_d(\boldsymbol{s}, y) = \sum_{\alpha\in\Gamma_d}\sum_{\beta\in\Gamma_d}\left(\sum_{\gamma\in\Gamma_{d_s}} s_\gamma y_{\alpha+\beta+\gamma}\right) E_{d,\alpha,\beta} = \sum_{\kappa\in\Gamma_{2d+d_s}} y_\kappa \bar{E}_{\boldsymbol{s},d,\kappa} \tag{6}$$

$$\forall\kappa \in \Gamma_{2d+d_s}:\ \bar{E}_{\boldsymbol{s},d,\kappa} = \sum_{\substack{\alpha\in\Gamma_d \\ \kappa-\alpha\in\Gamma_d}} \sum_{\substack{\gamma\in\Gamma_{d_s} \\ \kappa-\alpha-\gamma\in\Gamma_d}} s_\gamma E_{d,\alpha,\kappa-\alpha-\gamma} \tag{7}$$

Unconstrained polynomial problem

This part focuses on the unconstrained polynomial problem and its related notations and the other problems:

$$\min_{x\in\mathbb{R}^n} \boldsymbol{p_0}(x) \tag{8a}$$

Above, $\boldsymbol{p_0}(x)$ belongs to $P_{n,d_0}[x]$.

Based on problem (8), we define the following problem:

$$\min_{y\in\mathbb{R}^{|\Gamma_{2d_0}|}} p_0^T[y]_{d_0} \tag{9a}$$

$subject\ to$:

$$M_{d_0}(y) \geq 0 \tag{9b}$$

Problem (9) is convex and its dual problem is derived as follows:

$$\max_{X\in\mathbb{R}^{|\Gamma_{d_0}|\times|\Gamma_{d_0}|}} -trace\left(X\bar{E}_{d_0,0}\right) \tag{10a}$$

$subject\ to$:

$$\forall\gamma \in \Gamma_{2d_0}:\ trace\left(X\bar{E}_{d_0,\gamma}\right) = p_{0_\gamma} \tag{10b}$$

$$X \geq 0 \tag{10c}$$

Since problem (9) is convex, it satisfies strong duality condition (considering it has a strict feasible solution based on Slater's condition) which means $p_0^T[y^*]_{d_0} = -trace(X^*\bar{E}_0)$ where y^* and X^* are respectively the optimal solutions of problems (9) and (10).

Theorem 1. *Assume $\boldsymbol{p_0}(x) - \boldsymbol{p_0}(x^*)$ ($\boldsymbol{p_0}(0) = 0$) is a SOS polynomial where x^* is an optimal solution of problem (8). Then, the following equation will be satisfied:*

$$p_0^T[y^*]_{d_0} = \boldsymbol{p_0}(x^*) \tag{11}$$

Proof. Assume $\bar{y} = x^{*\{2d_0\}}$ that results in:

$$M_{d_0}(\bar{y}) = \sum_{\alpha\in\Gamma_{d_0}} \sum_{\beta\in\Gamma_{d_0}} x^{*\{d_0\}}_{\alpha+\beta} E_{d_0,\alpha,\beta} = \sum_{\alpha\in\Gamma_{d_0}} \sum_{\beta\in\Gamma_{d_0}} x^{*\alpha+\beta} E_{d_0,\alpha,\beta} \tag{12}$$

Simplifying (12), the following equations will be obtained:

$$M_{d_0}(\bar{y}) = \sum_{\alpha\in\Gamma_{d_0}} x^{*2\alpha} e_{d_0,\alpha} e_{d_0,\alpha}^T + \sum_{\alpha\in\Gamma_{d_0}} \sum_{\substack{\beta\in\Gamma_{d_0}\\ \beta\neq\alpha}} x^{*\alpha+\beta} \left(e_{d_0,\alpha} e_{d_0,\beta}^T + e_{d_0,\beta} e_{d_0,\alpha}^T\right) \tag{13}$$

$$\forall \alpha' \in \Gamma_{d_0}:\ \ e_{d_0,\alpha'}^T M_{d_0}(\bar{y}) e_{d_0,\alpha'} = x^{*2\alpha'} \tag{14}$$

$$\forall \alpha' \neq \beta' \in \Gamma_{d_0}:\ \ e_{d_0,\alpha'}^T M_{d_0}(\bar{y}) e_{d_0,\beta'} = x^{*\alpha'+\beta'} \tag{15}$$

Using (13-15), it can be successively concluded that $M_{d_0}(\bar{y}) = x^{*\{d_0\}} x^{*\{d_0\}^T} \geq 0$, $\bar{y}$ is a feasible solution of problem (9), and $p_0^T [y^*]_{d_0} \leq p_0^T [\bar{y}]_{d_0} = \boldsymbol{p_0}(x^*)$.

Due to $\boldsymbol{p_0}(x) - \boldsymbol{p_0}(x^*)$ is SOS, there exist polynomials $\left\{\boldsymbol{q_j}(x)\right\}_{j=1}^{r} \subset P_{d_0}[x]$ satisfying next relation:

$$\boldsymbol{p_0}(x) - \boldsymbol{p_0}(x^*) = \sum_{j=1}^{r} \boldsymbol{q_j}^2(x) \tag{16}$$

Now, define $\bar{X} = \sum_{j=1}^{r} q_j q_j^T$ that achieves next relations for arbitrary $x \in \mathbb{R}^n$:

$$trace(\bar{X}\bar{E}_0) = e_{d_0,0}^T \bar{X} e_{d_0,0}^T = \sum_{j=1}^{r} \left(q_j^T e_{d_0,0}\right)^2 = \sum_{j=1}^{r} \boldsymbol{q_j}^2(0) = -\boldsymbol{p_0}(x^*) \tag{17}$$

$$\boldsymbol{p_0}(x) - \boldsymbol{p_0}(x^*) = x^{\{d_0\}^T} \left(\sum_{j=1}^{r} q_j q_j^T\right) x^{\{d_0\}} = x^{\{d_0\}^T} \bar{X} x^{\{d_0\}} \tag{18}$$

$$x^{\{d_0\}^T} \bar{X} x^{\{d_0\}} = trace\left(\bar{X} x^{\{d_0\}} x^{\{d_0\}^T}\right) = trace\left(\bar{X} M_{d_0}\left(x^{\{d_0\}}\right)\right) \tag{19}$$

Simplifying (19), one has:

$$x^{\{d_0\}^T} \bar{X} x^{\{d_0\}} = trace\left(\bar{X}\bar{E}_{d_0,0}\right) + \sum_{\gamma\neq 0\in\Gamma_{2d_0}} x^{\gamma} trace\left(\bar{X}\bar{E}_{d_0,\gamma}\right) \tag{20}$$

Using (18) and (20), the following equation will be obtained:

$$\boldsymbol{p_0}(x) - \boldsymbol{p_0}(x^*) = -\boldsymbol{p_0}(x^*) + \sum_{\gamma\neq 0\in\Gamma_{2d_0}} p_{0_\gamma} x^{\gamma} = trace\left(\bar{X}\bar{E}_{d_0,0}\right) + \sum_{\gamma\neq 0\in\Gamma_{2d_0}} x^{\gamma} trace\left(\bar{X}\bar{E}_{d_0,\gamma}\right) \tag{21}$$

According to (21), we have:

$$\forall \gamma \in \Gamma_{2d_0}: \ trace\left(\bar{X}\bar{E}_{d_0,\gamma}\right) = p_{0_\gamma} \tag{22}$$

Since $\bar{X} = \sum_{j=1}^{r} q_j q_j^T \geq 0$ and holds (22), it can be simply concluded that $\bar{X}$ is a feasible solution of problem (10) that implies $trace\left(\bar{X}\bar{E}_{d_0,0}\right) \geq trace\left(X^*\bar{E}_{d_0,0}\right)$ and next relations:

$$trace\left(X^*\bar{E}_{d_0,0}\right) \leq trace\left(\bar{X}\bar{E}_{d_0,0}\right) = -\boldsymbol{p_0}(x^*) \tag{23}$$

$$trace\left(X^*\bar{E}_{d_0,0}\right) = -p_0^T[y^*]_{d_0} \geq -\boldsymbol{p_0}(x^*) \tag{24}$$

Regarding (23-24), we have $p_0^T[y^*]_{d_0} = -trace\left(X^*\bar{E}_{d_0,0}\right) = \boldsymbol{p_0}(x^*)$ that terminates the proof. □

In the sequel, the result of Theorem 1 has been extended to the general case that is proved by Theorem 2. For this purpose, it is needed to suppose the existence of sufficient large known parameter $b > 0$ satisgfying $\|x^*\| \leq b$. Also it is needed to define next problem:

Based on problem (8), we define the following problem for given $N \in \mathbb{Z}$:

$$\min_{y \in \mathbb{R}^{|\Gamma_{2N}|}} p_0^T[y]_{d_0} \tag{25a}$$

$subject\ to$:

$$M_N(y) \geq 0 \tag{25b}$$

$$M_{N-1}(\boldsymbol{t}, y) \geq 0 \tag{25c}$$

where t is the vector of coefficients of polynomial $\boldsymbol{t}(x) = b^2 - x^T x$.

Problem (25) is convex and its dual problem is derived as follows:

$$\max_{X \in \mathbb{R}^{|\Gamma_N| \times |\Gamma_N|}, Z \in \mathbb{R}^{|\Gamma_N| \times |\Gamma_N|}} -trace\left(X\bar{E}_{N,0}\right) - b^2 trace\left(Z\bar{E}_{N-1,0}\right) \tag{26a}$$

$subject\ to$:

$$\forall \gamma \in \Gamma_{2N}: \ trace\left(X\bar{E}_{N,\gamma}\right) + trace\left(Z\bar{E}_{\boldsymbol{t},N-1,\gamma}\right) = p_{0_\gamma} \tag{26b}$$

$$X, Z \geq 0 \tag{26c}$$

Theorem 2. *Assume* $\boldsymbol{p_0}(x) - \boldsymbol{p_0}(x^*)$ *(*$\boldsymbol{p_0}(0) = 0$*) is a PNS polynomial and* $\|x^*\| \leq b$ *for* $b > 0$*. Then, there exists sufficient large integer number* $N \in \mathbb{Z}_+$ *that satisfies next equation:*

$$p_0^T[y^*]_{d_0} = \boldsymbol{p_0}(x^*) \tag{27}$$

where y^* *is the optimal solution of problem (25) for* N*.*

Proof. Assume $\bar{y} = x^{*\{N\}}$ that yields:

$$M_N(\bar{y}) = \sum_{\alpha\in\Gamma_N}\sum_{\beta\in\Gamma_N} x^{*\{N\}}_{\alpha+\beta} E_{N,\alpha,\beta} = \sum_{\alpha\in\Gamma_N}\sum_{\beta\in\Gamma_N} x^{*\alpha+\beta} E_{N,\alpha,\beta} \tag{28}$$

$$M_{N-1}(\boldsymbol{t},\bar{y}) = \sum_{\alpha\in\Gamma_{N-1}}\sum_{\beta\in\Gamma_{N-1}}\left(\sum_{\gamma\in\Gamma_2} t_\gamma x^{*\{N\}}_{\alpha+\beta+\gamma}\right) E_{N-1,\alpha,\beta} =$$

$$\sum_{\alpha\in\Gamma_{N-1}}\sum_{\beta\in\Gamma_{N-1}}\left(\sum_{\gamma\in\Gamma_2} t_\gamma x^{*\alpha+\beta+\gamma}\right) E_{N-1,\alpha,\beta} \tag{29}$$

Simplifying (28-29), the following equations will be obtained:

$$M_N(\bar{y}) = \sum_{\alpha\in\Gamma_N} x^{*2\alpha} e_{N,\alpha} e_{N,\alpha}^T + \sum_{\alpha\in\Gamma_N}\sum_{\substack{\beta\in\Gamma_N\\ \beta\neq\alpha}} x^{*\alpha+\beta}\left(e_{N,\alpha} e_{N,\beta}^T + e_{N,\beta} e_{N,\alpha}^T\right) \tag{30}$$

$$M_{N-1}(\boldsymbol{t},\bar{y}) = \sum_{\alpha\in\Gamma_{N-1}}\left(\sum_{\gamma\in\Gamma_2} t_\gamma x^{*2\alpha+\gamma}\right) e_{N-1,\alpha} e_{N-1,\alpha}^T +$$

$$\sum_{\alpha\in\Gamma_{N-1}}\sum_{\beta\in\Gamma_{N-1}}\left(\sum_{\gamma\in\Gamma_2} t_\gamma x^{*\alpha+\beta+\gamma}\right)\left(e_{N-1,\alpha} e_{N-1,\beta}^T + e_{N-1,\beta} e_{N-1,\alpha}^T\right) \tag{31}$$

$$\forall \alpha' \in \Gamma_{N-1}:\ \ e_{N-1,\alpha'}^T M_{N-1}(\boldsymbol{t},\bar{y}) e_{N-1,\alpha'} = \boldsymbol{t}(x^*) x^{*2\alpha'} \tag{32}$$

$$\forall \alpha' \neq \beta' \in \Gamma_{N-1}:\ \ e_{N-1,\alpha'}^T M_{N-1}(\boldsymbol{t},\bar{y}) e_{N-1,\beta'} = \boldsymbol{t}(x^*) x^{*\alpha'+\beta'} \tag{33}$$

Equations (30-33) achieves $M_N(\bar{y}) = x^{*\{N\}} x^{*\{N\}^T} \geq 0$ and $M_{N-1}(\boldsymbol{t},\bar{y}) = \boldsymbol{t}(x^*) x^{*\{N-1\}} x^{*\{N-1\}^T} \geq 0$ that implies $\bar{y}$ is a feasible solution of problem (25), and $p_0^T[y^*]_{d_0} \leq p_0^T[\bar{y}]_{d_0} = \boldsymbol{p_0}(x^*)$.

Due to $\boldsymbol{p_0}(x) - \boldsymbol{p_0}(x^*)$ is PNS, there exist polynomials $\{\boldsymbol{q_j}(x)\}_{j=1}^{r_1} \subset P_N[x]$ and $\{\boldsymbol{s_j}(x)\}_{j=1}^{r_2} \subset P_N[x]$ satisfying next relation for each $\varepsilon > 0$:

$$\boldsymbol{p_0}(x) - \boldsymbol{p_0}(x^*) + \varepsilon = \sum_{j=1}^{r_1} \boldsymbol{q_j}^2(x) + \boldsymbol{t}(x)\sum_{j=1}^{r_2} \boldsymbol{s_j}^2(x) \tag{34}$$

Now, define $\bar{X} = \sum_{j=1}^{r_1} q_j q_j^T \in \mathbb{R}^{|\Gamma_N|\times|\Gamma_N|}$ and $\bar{Z} = \sum_{j=1}^{r_2} s_j s_j^T \in \mathbb{R}^{|\Gamma_{N-1}|\times|\Gamma_{N-1}|}$ that achieves next relations for arbitrary $x \in \mathbb{R}^n$:

$$trace(\bar{X}\bar{E}_0) + b^2 trace\left(Z\bar{E}_{N-1,0}\right) = e_{N,0}^T \bar{X} e_{N,0}^T + b^2 e_{N-1,0}^T \bar{Z} e_{N-1,0}^T = \sum_{j=1}^{r_1}\left(q_j^T e_{N,0}\right)^2 + b^2 \sum_{j=1}^{r_2}\left(s_j^T e_{N-1,0}\right)^2 = \sum_{j=1}^{r_1} \boldsymbol{q_j}^2(0) + \boldsymbol{t}(0) \sum_{j=1}^{r_2} \boldsymbol{s_j}^2(0) = -\boldsymbol{p_0}(x^*) + \varepsilon \tag{35}$$

$$\boldsymbol{p_0}(x) - \boldsymbol{p_0}(x^*) + \varepsilon = x^{\{N\}^T} \bar{X} x^{\{N\}} + \boldsymbol{t}(x) x^{\{N-1\}^T} \bar{Z} x^{\{N-1\}} \tag{36}$$

$$x^{\{N\}^T} \bar{X} x^{\{N\}} + \boldsymbol{t}(x) x^{\{N-1\}^T} \bar{Z} x^{\{N-1\}} = trace\left(\bar{X} M_N\left(x^{\{N\}}\right)\right) + trace\left(\bar{Z} M_{N-1}\left(\boldsymbol{t}, x^{\{N-1\}}\right)\right) \tag{37}$$

Simplifying (37), one has:

$$x^{\{N\}^T} \bar{X} x^{\{N\}} + \boldsymbol{t}(x) x^{\{N-1\}^T} \bar{Z} x^{\{N-1\}} = trace\left(\bar{X}\bar{E}_{N,0}\right) + \sum_{\gamma\neq 0\in\Gamma_{2N}} x^\gamma trace\left(\bar{X}\bar{E}_{N,\gamma}\right) + trace\left(\bar{Z}\bar{E}_{\boldsymbol{t},N-1,0}\right) + \sum_{\gamma\neq 0\in\Gamma_{2N}} x^\gamma trace\left(\bar{Z}\bar{E}_{\boldsymbol{t},N-1,\gamma}\right) \tag{38}$$

$$\boldsymbol{p_0}(x) - \boldsymbol{p_0}(x^*) + \varepsilon = -\boldsymbol{p_0}(x^*) + \sum_{\gamma\neq 0\in\Gamma_{2N}} p_{0_\gamma} x^\gamma = \left(trace\left(\bar{X}\bar{E}_{N,0}\right) + trace\left(\bar{Z}\bar{E}_{\boldsymbol{t},N-1,0}\right)\right) + \sum_{\gamma\neq 0\in\Gamma_{2N}} x^\gamma \left(trace\left(\bar{X}\bar{E}_{N,\gamma}\right) + trace\left(\bar{Z}\bar{E}_{\boldsymbol{t},N-1,\gamma}\right)\right) \tag{39}$$

According to (39), we have:

$$\forall\gamma \in \Gamma_{2N}:\ \ trace\left(\bar{X}\bar{E}_{N,\gamma}\right) + trace\left(\bar{Z}\bar{E}_{\boldsymbol{t},N-1,\gamma}\right) = p_{0_\gamma} \tag{40}$$

Since $\bar{X} = \sum_{j=1}^{r} q_j q_j^T \geq 0$, $\bar{Z} = \sum_{j=1}^{r_2} s_j s_j^T \geq 0$ and they holds (40), it can be simply concluded that pair $(\bar{X}, \bar{Z})$ belongs to the feasible region of problem (26) that implies $trace\left(\bar{X}\bar{E}_{N,0}\right) + b^2 trace\left(\bar{Z}\bar{E}_{\boldsymbol{t},N-1,0}\right) \geq trace\left(X^*\bar{E}_{N,0}\right) + b^2 trace\left(Z^*\bar{E}_{\boldsymbol{t},N-1,0}\right)$ and next relations:

$$trace\left(X^*\bar{E}_{N,0}\right) + b^2 trace\left(Z^*\bar{E}_{\boldsymbol{t},N-1,0}\right) \leq trace\left(\bar{X}\bar{E}_{N,0}\right) + b^2 trace\left(\bar{Z}\bar{E}_{\boldsymbol{t},N-1,0}\right) = -\boldsymbol{p_0}(x^*) + \varepsilon \tag{41}$$

$$trace\left(X^*\bar{E}_{N-1,0}\right) + b^2 trace\left(Z^*\bar{E}_{\boldsymbol{t},N-1,0}\right) = -p_0^T [y^*]_N \geq -\boldsymbol{p_0}(x^*) + \varepsilon \tag{42}$$

Regarding (41-42), we have $\boldsymbol{p_0}(x^*) - \varepsilon \leq p_0^T[y^*]_N \leq \boldsymbol{p_0}(x^*)$ that terminates the proof. □

In proofs of the above theorems, some major properties are repeatedly addressed that are given below:

$$\forall x \in \mathbb{R}^n: \quad M_d\left(x^{\{d\}}\right) = x^{\{d\}}{x^{\{d\}}}^T \tag{43}$$

$$\forall x \in \mathbb{R}^n: \quad M_{d-\frac{d_s}{2}}\left(\boldsymbol{s}, x^{\{d\}}\right) = \boldsymbol{s}(x)x^{\left\{d-\frac{d_s}{2}\right\}}{x^{\left\{d-\frac{d_s}{2}\right\}}}^T \tag{44}$$

Constrained polynomial problem

Firstly, consider the following constrained polynomial problem:

$$\min_{x\in\mathbb{R}^n} \boldsymbol{p_0}(x) \tag{45a}$$

$subject\ to$:

$$\forall i \in \{1, \dots, m\}: \ \boldsymbol{p_i}(x) \geq 0 \tag{45b}$$

Above, $\boldsymbol{p_i}(x)$ belongs to $P_{d_i}[x]$ for all $i \in \{1, \dots, m\}$. Based on (45), the following LMI problem is reached for $N \in \mathbb{Z}_+$:

$$\min_{y\in\mathbb{R}^{|\Gamma_N|}} p_0^T[y]_{d_0} \tag{46a}$$

$subject\ to$:

$$M_N(y) \geq 0 \tag{46b}$$

$$\forall i \in \{1, \dots, m\}: \ M_{N-\frac{d_i}{2}}(\boldsymbol{p_i}, y) \geq 0 \tag{46c}$$

The dual problem of (46) will be obtained as follows:

$$\max_{X,\{Z_i\}_{i=1}^m} -trace\left(X\bar{E}_{N,0}\right) - \sum_{i=1}^m \boldsymbol{p_i}(0)trace\left(Z_i\bar{E}_{N-\frac{d_i}{2},0}\right) \tag{47a}$$

$subject\ to$:

$$\forall \gamma \in \Gamma_{2N}: \ trace\left(X\bar{E}_{N,\gamma}\right) + \sum_{i=1}^m trace\left(Z_i\bar{E}_{\boldsymbol{p_i},N-\frac{d_i}{2},\gamma}\right) = p_{0_\gamma} \tag{47b}$$

$$X = \mathbb{R}^{|\Gamma_N|\times|\Gamma_N|} \geq 0 \tag{47c}$$

$$\forall i \in \{1, \dots, m\}: \; Z_i = \mathbb{R}^{\left|\Gamma_{N-\frac{d_i}{2}}\right| \times \left|\Gamma_{N-\frac{d_i}{2}}\right|} \geq 0 \tag{47d}$$

Theorem 3. *Assume x^* is the optimal solution of problem (45). Then, there exists a sufficient large integer number $N \in \mathbb{Z}_+$ that satisfies the next equation:*

$$p_0^T [y^*]_N = \boldsymbol{p_0}(x^*) \tag{48}$$

where y^* is the optimal solution of problem (46) for N.

Proof. Suppose $\bar{y} = x^{*\{N\}} x^{*\{N\}^T}$ that implies the next relations:

$$M_N(\bar{y}) = x^{*\{N\}} x^{*\{N\}^T} \geq 0 \tag{49}$$

$$\forall i \in \{1, \dots, m\}: \; M_{N-\frac{d_i}{2}}(\boldsymbol{p_i}, \bar{y}) = \boldsymbol{p_i}(x^*) x^{*\{N\}} x^{*\{N\}^T} \geq 0 \tag{50}$$

Due to (49-50), we have $p_0^T [\bar{y}]_N \geq p_0^T [y^*]_N$.

Since x^* is the optimal solution to problem (45), it can be simply concluded that polynomial $\boldsymbol{p_0}(x) - \boldsymbol{p_0}(x^*)$ is at least a positive polynomial over the semi-algebraic feasible region of this problem. Through this fact, there exists sufficient large positive integer number N for each $\varepsilon > 0$ satisfying the next relation:

$$\boldsymbol{p_0}(x) - \boldsymbol{p_0}(x^*) + \varepsilon = \sum_{k=1}^{r_0} \boldsymbol{q_k}^2(x) + \sum_{i=1}^{m} \boldsymbol{p_i}(x) \sum_{k=1}^{r_i} \boldsymbol{s_{i,k}}^2(x) \tag{51}$$

where $\{\boldsymbol{q_k}(x)\}_{k=1}^{r_0} \subset P_N[x]$ and $\left\{\boldsymbol{s_{i,k}}(x)\right\}_{k=1}^{r_i} \subset P_{N-d_i}[x]$.

Based on (51), define next symmetric positive-semi-definite matrices:

$$\bar{X} = \sum_{k=1}^{r_0} q_k q_k^T \tag{52}$$

$$\forall i \in \{1, \dots, m\}: \; \bar{Z}_i = \sum_{k=1}^{r_i} s_{i,k} s_{i,k}^T \tag{53}$$

Using the above definitions, we have:

$$\boldsymbol{p_0}(x) - \boldsymbol{p_0}(x^*) + \varepsilon = x^{\{N\}^T} \bar{X} x^{\{N\}} + \sum_{i=1}^{m} \boldsymbol{p_i}(x) x^{\left\{N-\frac{d_i}{2}\right\}^T} \bar{Z}_i x^{\left\{N-\frac{d_i}{2}\right\}} \tag{54}$$

$$\boldsymbol{p_0}(x) - \boldsymbol{p_0}(x^*) + \varepsilon = trace\left(\bar{X} M_N\left(x^{\{N\}}\right)\right) +$$

$$\sum_{i=1}^{m} trace\left(\bar{Z}_i M_{N-\frac{d_i}{2}}\left(\boldsymbol{p_i}, x^{\left\{N-\frac{d_i}{2}\right\}}\right)\right) \tag{55}$$

$$trace\left(\bar{X}M_N\left(x^{\{N\}}\right)\right)+\sum_{i=1}^{m} trace\left(\bar{Z}_i M_{N-\frac{d_i}{2}}\left(\boldsymbol{p_i}, x^{\left\{N-\frac{d_i}{2}\right\}}\right)\right) \quad =$$

$$\sum_{\gamma\in\Gamma_{2N}}\left(trace\left(\bar{X}\bar{E}_{N,\gamma}\right)+\sum_{i=1}^{m} trace\left(\bar{Z}_i\bar{E}_{\boldsymbol{p_i},N-\frac{d_i}{2},\gamma}\right)\right)x^{\gamma} \tag{56}$$

$$\forall\gamma\neq 0\in\Gamma_{2N}:\ \ trace\left(\bar{X}\bar{E}_{N,\gamma}\right)+\sum_{i=1}^{m} trace\left(\bar{Z}_i\bar{E}_{\boldsymbol{p_i},N-\frac{d_i}{2},\gamma}\right)=p_{0_\gamma} \tag{57}$$

$$trace\left(X\bar{E}_{N,0}\right)+\sum_{i=1}^{m}\boldsymbol{p_i}(0)trace\left(Z_i\bar{E}_{N-\frac{d_i}{2},0}\right)=\boldsymbol{p_0}(x^*)-\varepsilon \tag{58}$$

Regarding (54-58), $(\bar{X},\{\bar{Z}_i\}_{i=1}^{m})$ is a feasible solution of the dual of dual problem. Thus, we have:

$$\boldsymbol{p_0}(x^*)-\varepsilon\leq p_0^T[y^*]_N\leq\boldsymbol{p_0}(x^*) \tag{59}$$

The above relation terminates the proof.